

Unit 1.1. Numbers, Intervals, and Inequalities.

Source: Stewart, Appendix A pp A2 - A6.

Learning objectives

On completion of this unit you should

1. know the definitions of the following number systems: the set $\mathbb{N}$ of all natural numbers, the set $\mathbb{Z}$ of all integers, the set $\mathbb{Q}$ of all rational numbers, the set $\mathbb{I}$ of all irrational numbers and the set $\mathbb{R}$ of all real numbers;
2. be able to use set notation;
3. be able to use the word "and" for the intersection of two sets;
4. be able to use the word "or" for the union of two sets;
5. be able to find the union/intersection of two sets;
6. be able to use interval notation;
7. know the rules for inequalities and be able to use it;
8. be able to solve inequalities.

Numbers

Integers ($\mathbb{Z}$); $\dots -2, -1, 0, 1, 2, 3, \dots$

Rational numbers ($\mathbb{Q}$); ratio of integers i.e. any rational no. 'r' can be expressed as $r = \frac{m}{n}$, where m & n are integers, & $n \neq 0$.

Numbers with repeating decimals are also rational numbers.
E.g. $\frac{1}{2}$; $3 = \frac{3}{1}$; $\frac{2}{3} = 0.66\bar{6}$, $\frac{2}{99} = 0.0202\bar{02}$

Irrational numbers $\mathbb{I}$; numbers that Can't be expressed as a ratio of integers. e.g. $\sqrt{2}$, π , $\sqrt{3}$

Intervals

Geometrically, Intervals corresponds to line segments.

E.g.



Interval notation In general, $x \in (a, b) = \{x \mid a < x < b\}$ set builder notation

$$x \in [a, b] = \{x \mid a \leq x \leq b\}$$

Infinite intervals ; $x \in (a, \infty)$; $x \in (-\infty, a)$

Combining Intervals

- Union $\cup$ (OR)
- Intersection $\cap$ (AND)

Examples

1 Write the following in interval notation

a) $x > 3$ and $x < 5$

b) $x < 3$ or $x \geq 5$

c) $x < 0$ and $x > 5$

2) Solve for x ;

a) $x^2 - 1 = 0$

b) $x(x-1) = 0$

Inequalities.

Rules of inequalities

1. If $a < b$, then $a + c < b + c$
2. If $a < b$ and $c < d$, then $a + c < b + d$
3. If $a < b$ and $c > 0$, then $ac < bc$
4. If $a < b$ and $c < 0$, then $ac > bc$
5. If $0 < a < b$, then $\frac{1}{a} > \frac{1}{b}$

Examples

3 Solve for x , and write your answer in interval notation

a) $1 + x < 7x + 5$

b) $(x-1)(x-2) > 0$

c) $x^3 + 3x < 4x^2$

d) $abc + b < c$ if $a, b, c < 0$

e) $\frac{x+1}{2-x} \geq 0$

f) $\frac{1}{x+1} > \frac{1}{x+2}$